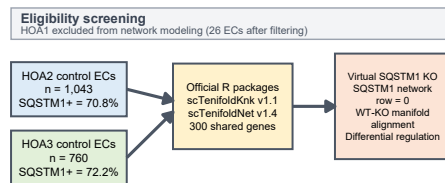


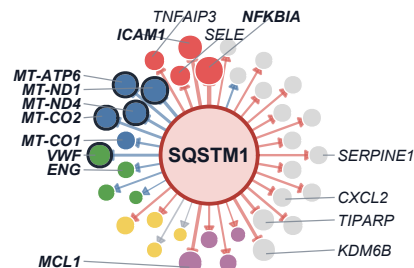
A Official-package virtual-knockout design



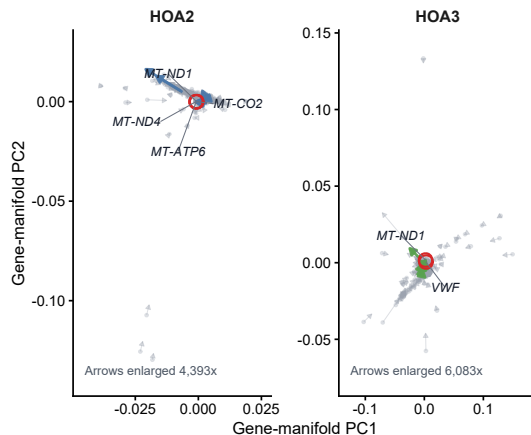
Parameter audit

Primary: 20 networks, $q=0.95$, 30 manifold dimensions
 Package-default sensitivity: 10 networks, $q=0.90$, 2 dimensions
 Calibration: 295-gene mtDNA-feature-excluded refits + 20 matched-comparator KOs (Table S10)
 No downstream gene or pathway was FDR-significant in both donors under both profiles.

B SQSTM1 WT neighborhood removed by virtual KO

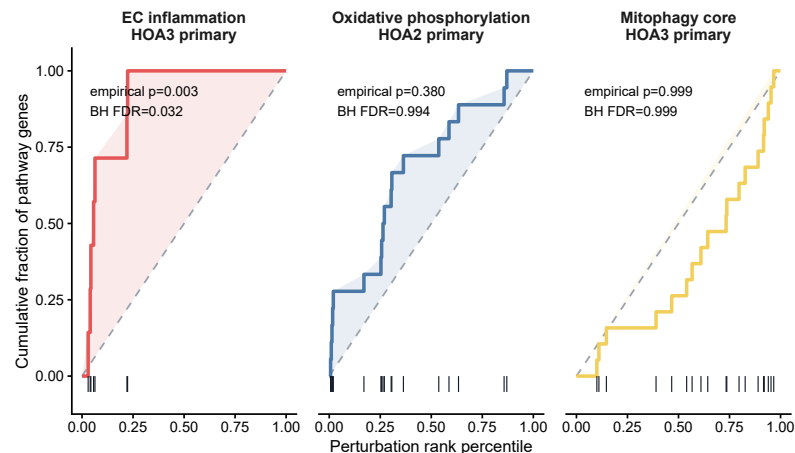


C WT-to-virtual-KO gene-manifold displacement



Gray: downstream genes; color: top 10 ranks; red rings: FDR<0.05.
 Arrows are uniformly enlarged gene-manifold displacements.

D Matched-null pathway-rank recovery



Pathway-member recovery along official perturbation ranks.
 p /FDR use 20,000 expression- and prevalence-matched null sets.